

10 (**long**) In a proposed fairground ride, an unpowered trolley rolls, without friction, along a U-shaped track, as illustrated in Fig. 6. The height of the track has the form $h = ks^4$, where s is the distance along the track measured from O.

The trolley is released at rest from $s = s_0$, and there is a gravitational field g .

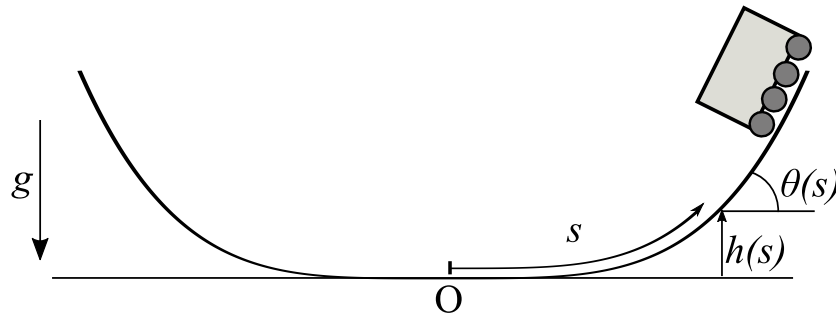


Fig. 6

- (a) Show that the track makes an angle with the horizontal $\theta(s) = \sin^{-1} \left(\frac{dh}{ds} \right)$. [2]
- (b) Find the velocity of the trolley as a function of s . [4]
- (c) Find the tangential acceleration of the trolley as a function of s . [3]
- (d) Using the approximation $\sin \theta \approx \theta$, find the acceleration of the trolley normal to the track as a function of s . [5]
- (e) Find the locations, s , on the track where the trolley's velocity, tangential acceleration and normal acceleration are maximum. For normal acceleration you may still assume $\sin \theta \approx \theta$. [8]
- (f) (i) The trolley returns to its original position a time T after it is released. Explain why T can only depend on g and k via their product, gk . [2]
 - (ii) If the trolley is released at $s_0 = 1$ m, it takes 1 second to return to its original position. Use dimensional analysis to calculate the return time if $s_0 = 5$ m. [6]